MIDAS: An Energy-Efficient Microscaling Quantization-Based Digital CIM Accelerator With Spatio-Temporal Cyclic Alignment for Generative AI

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Abstract*—Digital compute-in-memory (DCIM) and the microscaling (MX) data format have emerged as promising hardware and algorithmic solutions, respectively, for energy-efficient generative AI (GenAI) processing. However, their combination introduces challenges, including redundant computation from mantissa alignment, excessive exponent handling overhead, and accuracy loss from static approximation schemes that fail to accommodate block-wise dynamic bit-significance. In this article, we present MIDAS, an MX-optimized DCIM-based GenAI accelerator with three key features: 1) spatio-temporal cyclic alignment (STCA) with sign-magnitude-based assembled column addition (SACA) enables compact weight mapping, high input reusability, and low-latency computation, achieving 1.72× higher macro energy efficiency; 2) MX-aware minimum exponent post-detection (MEPD) eliminates redundant maximum exponent detection by exploiting the inherent characteristics of MX quantization, reducing exponent handling power, area, and latency by 68.8%, 63.2%, and 22.5%, respectively; and 3) sense amplifier-based dynamic bit (SADB)-significance handling with shared scale-aware channel regrouping (SSCR) dynamically adjusts approximation bit-positions according to each block's shared scale, improving energy efficiency by 27.3% while minimizing accuracy loss. MIDAS is fabricated in Samsung 28-nm CMOS technology and occupies a 7.56 mm^2 die area. MIDAS achieves macro energy and area efficiencies of 96.3 TFLOPS/W and 0.69 TFLOPS/mm^2, respectively, under MXFP4. Compared to state-of-the-art DCIMs, MIDAS demonstrates 1.58× higher system energy efficiency and 3.32× higher macro energy efficiency, validating its effectiveness as an MX-optimized AI accelerator for next-generation GenAI applications.

***Index Terms*—Compute-in-memory (CIM), energy-efficient, generative artificial intelligence (GenAI), microscaling (MX) format.**

Received 27 January 2026; revised 2 April 2026; accepted 28 April 2026. Date of publication 22 May 2026; date of current version 25 September 2026. This article was approved by Associate Editor Kyuho Jason Lee. This work was supported by Samsung Electronics Co., Ltd., under Grant IO251215-14633-01. *(Corresponding author: Hoi-Jun Yoo.)*

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Digital Object Identifier 10.1109/JSSC.2026.3692361

I. Introduction

GENERATIVE AI (GenAI), especially large-language models (LLMs), has demonstrated significant performance improvements, leading to its widespread adoption. However, these models involve an exponential increase in parameter count and computational complexity, resulting in substantial memory transactions and high computational costs. Hence, deploying GenAI services on resource-constrained edge devices remains a critical challenge for modern computing systems.

To address these bottlenecks, research has focused on overcoming the "Memory Wall" inherent in traditional von Neumann architectures. Compute-in-memory (CIM) has emerged as a promising solution, performing operations directly within the memory array to minimize data movement and maximize energy efficiency. Unlike analog CIM, which is susceptible to signal distortion and accuracy loss due to process, voltage, and temperature (PVT) variations, digital CIM (DCIM) leverages digital logic to achieve high energy efficiency without sacrificing computational precision [1], [2], [3], [4], [5], [6]. These characteristics make DCIM well-suited for processing accuracy-sensitive GenAI models.

In addition to hardware-level optimizations, low-bit quantization and specialized data formats are widely used to reduce data size and bandwidth requirements. However, previous custom quantization methods have suffered from limited generalizability and complex software overhead [7]. To resolve these issues, a consortium of global industry-leading companies —AMD, Arm, Intel, Meta, Microsoft, NVIDIA, and Qualcomm—has collaboratively standardized the microscaling (MX) data format [8], [9]. As shown in Fig. 1(a), the MX format extends the block floating point (BFP) concept by grouping elements with a shared scale, maintaining high accuracy while reducing hardware complexity. Among MX-compliant formats, MXFP allows each element within a block to retain its own exponent bits, providing additional scaling capability that effectively handles intra-block variance. This feature enables MXFP to achieve strong performance on various LLM tasks, even at sub-8-bit precision as shown

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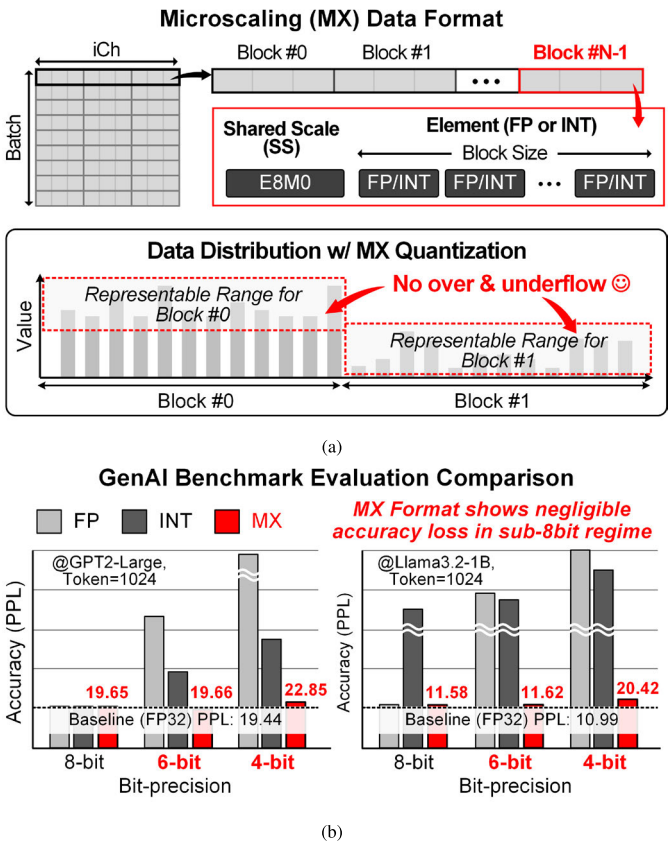


Fig. 1. (a) MX data format and its quantization range and (b) GenAI benchmark evaluation.

in Fig. 1(b). While CIM and the MX format each offer clear benefits for energy-efficient GenAI processing, their combination introduces challenges that existing DCIMs have yet to address, as illustrated in Fig. 2. The primary challenge arises from mantissa alignment, which results in redundant computations and severe cell underutilization. While this issue is inherent in prior floating-point (FP) CIMs, it becomes more severe in the MX format. Traditional designs typically rely on mantissa truncation to mitigate this overhead. However, such a strategy is no longer viable in the sub-8-bit regime of MX, where even minimal truncation induces severe accuracy degradation. The second challenge involves the excessive overhead of exponent handling logic. Operations such as maximum exponent searching and activation mantissa alignment account for an increasing portion of energy and area as the bit width decreases. Specifically, the energy-area product for exponent handling in MXFP4 is 45.4 percentage points higher than in FP16, identifying exponent operations as a critical bottleneck in low-bit MX processing. The third challenge stems from the disparate bit-significances encountered during MX-quantized GEMM operations, induced by varying shared scales across blocks. Prior DCIM architectures [1], [5], [10], [11] commonly employ approximation techniques for energy-efficient computation, replacing full adders (FAs) with low-cost OR gates at lower bit-positions. Such techniques yield significant power savings with minimal accuracy loss in conventional fixed-point and FP operations that utilize a single-scale factor per tensor. However, since these designs statically assign the bit-positions for approximation, they fail to accommodate the dynamic significance shifts inherent in block-wise scaling.

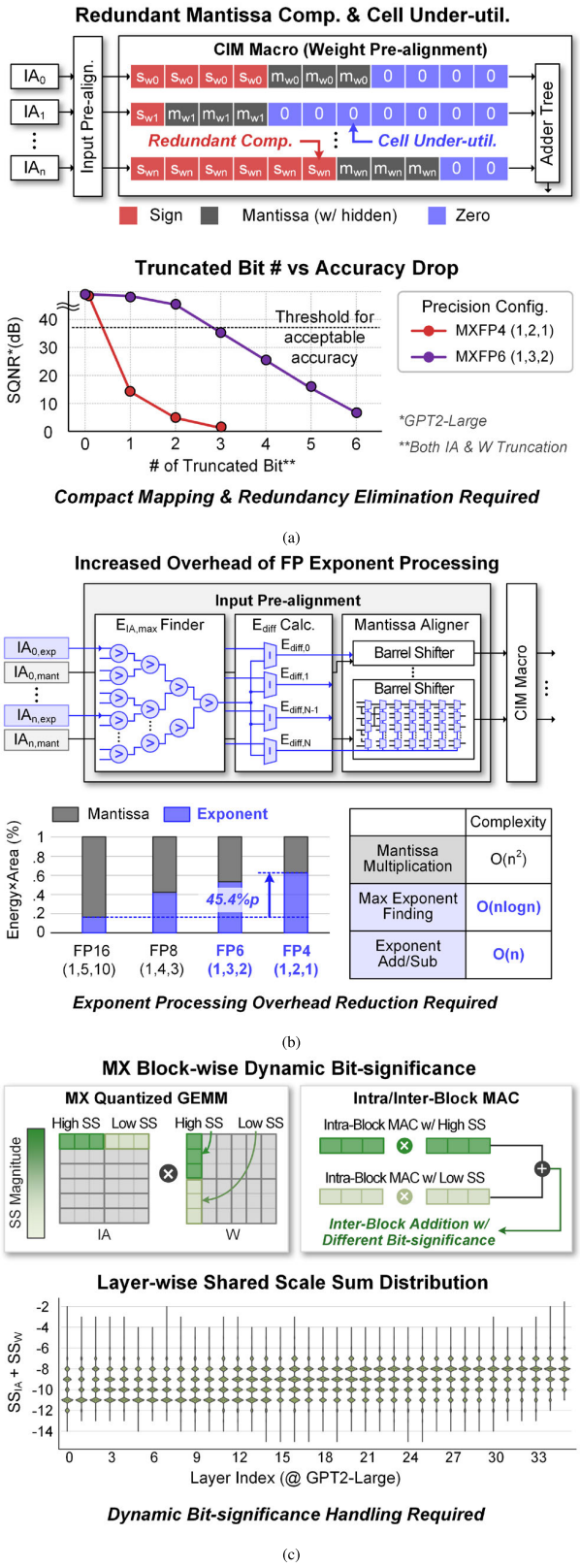


Fig. 2. Hardware challenges of MX-supported DCIM. (a) Redundant mantissa computation and cell under-utilization. (b) Increased overhead of FP exponent processing. (c) MX block-wise dynamic bit-significance.

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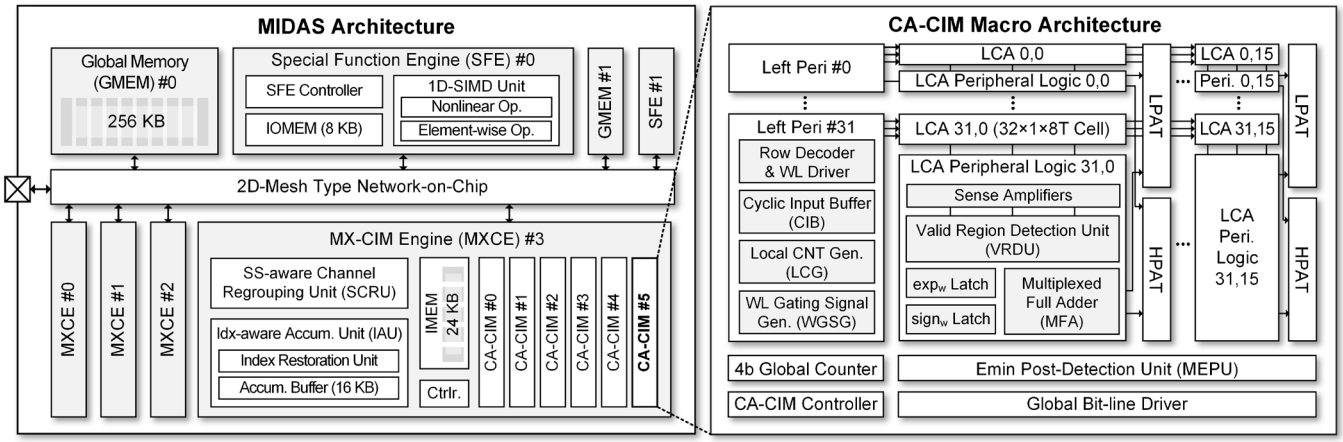


Fig. 3. Overall chip architecture of the proposed MIDAS.

As a result, significant accuracy loss occurs during inter-block addition in MX-quantized GEMM. To address these challenges, we propose MIDAS, an MX-optimized DCIM-based GenAI accelerator that achieves high energy and area efficiency. The main contributions are summarized as follows.

1) Spatio-temporal cyclic alignment (STCA) facilitates high input reusability and compact weight mapping by leveraging temporally cyclic alignment for input activations and spatially cyclic alignment for weights.
2) Sign-magnitude-based assembled column addition (SACA) performs partial product (PP) accumulation within the intra-LCA using sign-magnitude representation. Additionally, it employs assembled column addition, which shifts low-bit-significance (LBS) PP columns toward high-bit-significance (HBS) PP columns, enabling low-latency computation.
3) MX-aware minimum exponent post-detection (MEPD) demonstrates that maximum exponent detection, commonly used in conventional FP operations, is redundant in the MX format. In its place, it introduces a hardware-efficient methodology for minimum exponent detection, providing an effective, low-cost alternative for exponent handling.
4) Sense amplifier-based dynamic bit (SADB)-significance handling enables selection of the sense amplifier reference voltage, allowing the bit-position for approximation to be dynamically adjusted according to each block's shared scale. This flexibility ensures high energy efficiency even across blocks with varying shared scales.
5) Shared scale-aware channel regrouping (SSCR) further augments the energy efficiency gains of approximation by regrouping channels so that blocks with similar shared scales are mapped to the same CIM macro.

The remainder of this article is organized as follows. Section II provides an overview of the chip architecture. Section III details the proposed mantissa alignment method and dataflow tailored to the MX format. Section IV elaborates on the proposed MXFP processing pipeline and its related circuits. Section V describes the proposed SADB-significance handling and shared scale (SS) regrouping scheme with its related hardware units. Section VI presents the measurement results, and Section VII concludes this article.

II. MIDAS Architecture

Fig. 3 illustrates the overall chip architecture of MIDAS. MIDAS comprises four MX-CIM engines (MXCEs), two global memories (GMEMs), and two special function engines (SFEs), all interconnected via a 2-D-mesh network-on-chip (NoC). Each MXCE consists of a shared-scale-aware channel regrouping unit (SCRU), an index-aware accumulation unit (IAU), 24 KB of input memory (IMEM), and six 16 Kb cyclic alignment-based CIM (CA-CIM) macros for weight-stationary computation. The detailed operation and implementation of SCRU and IAU will be explained in Section V. The CA-CIM macro contains a 4-bit global counter, a minimum exponent post-detection unit (MEPU), a global bitline (BL) driver, a CA-CIM controller, 32 left peripheral logic blocks, and 16 local cell array (LCA) columns. The left peripheral logic includes a row decoder, a word-line (WL) driver, a cyclic input buffer (CIB), a local counter generator (LCG), and a WL gating signal generator (WGSG). Each LCA column comprises 32 LCAs, each with its peripheral logic, an LBS PP adder tree (LPAT), and an HBS PP adder tree (HPAT). The LCA peripheral logic comprises sense amplifiers for each BL, a valid region detection unit (VRDU), a weight exponent latch, a weight sign latch, and a multiplexed FA (MFA). Each LCA contains 32 8T SRAM cells connected to three WLs and three BLs. The bitwise multiplication results read from the three BLs are simultaneously added in the LCA peripheral logic and forwarded to the LPAT and HPAT. The SFE comprises an SFE controller, 8 KB of input/output memory (IOMEM), and a 1D-SIMD unit for nonlinear and element-wise operations. The GMEM consists of 256 KB of SRAM. The two GMEMs are primarily designed for storing intermediate feature maps rather than weights. GMEMs enable intermediate feature maps from one layer to be directly reused by the subsequent layer without being transferred off-chip, thereby reducing external memory access.

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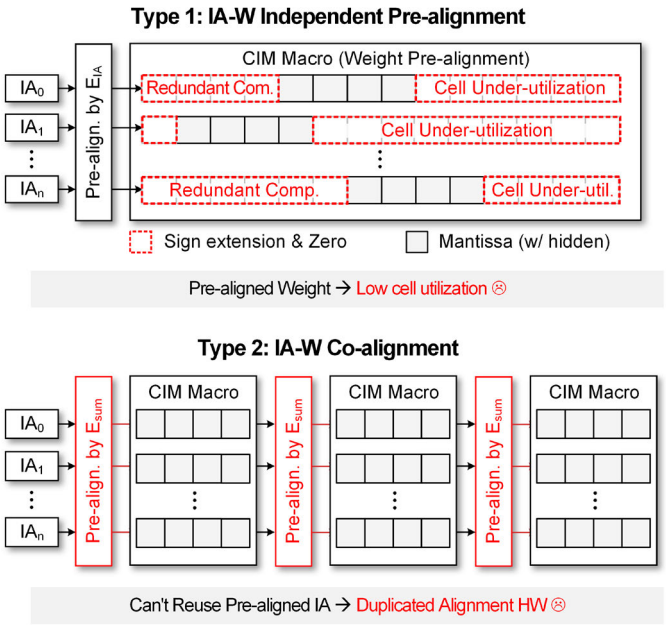


Fig. 4. Limitations of previous mantissa alignment methods.

III. Mantissa Alignment Methodology and Dataflow of MIDAS

A. Limitations of Previous Mantissa Alignment Methods

FP-supporting CIMs typically employ a mantissa pre-alignment scheme that incorporates exponent values into the mantissa before multiplication, rather than a post-alignment scheme that reflects exponent values after mantissa multiplication before sending results to the adder tree. This approach reduces the overhead of adder trees, which constitute a significant portion of area and power consumption [3], [5], [6]. As shown in Fig. 4, the pre-alignment schemes are categorized into two types.

1) Input Activation-Weight Independent Pre-Alignment: The first type independently pre-aligns inputs and weights based on their respective exponents [1], [2], [3], [10], [11], [12], [13]. Since input pre-alignment has no dependency on weight exponents, this method enables the reuse of pre-aligned inputs across multiple output channel weights. However, because weights are stored in the cell array in a pre-aligned state, redundant computation arises from sign extension at the MSB and zero padding at the LSB, resulting in wasted cells. Previous approaches have addressed this redundancy by truncating weights after pre-alignment. However, in the sub-8-bit regime targeted by the MX format, truncation can severely degrade accuracy on generative AI (GenAI) tasks.

2) Input Activation-Weight Co-Alignment: The second type pre-aligns inputs based on the sum of input and weight exponents [4], [10], [14]. Since weights are not stored in a pre-aligned form, this method achieves high cell utilization. However, pre-aligned inputs cannot be reused across output channels, requiring duplicated alignment logic for each output channel weight and increasing hardware cost. As exponent processing accounts for a larger portion of hardware cost at lower bit precisions, this problem becomes more pronounced in the sub-8-bit MX format. To address both the redundant computation and cell utilization issues in type 1 and the duplicated alignment logic problem in type 2, we propose a new mantissa alignment method called spatio-temporal cyclic alignment.

B. Spatio-Temporal Cyclic Alignment

As depicted in Fig. 5, the key idea of STCA is to apply temporal cyclic alignment to inputs and spatial cyclic alignment to weights. Each input determines its initial bit-feeding order based on its exponent and is then cyclically shifted by 1 bit per cycle. Meanwhile, weights are cyclically shifted offline based on their exponent values before being stored in memory. Because weight alignment is performed cyclically, high cell utilization is achieved. Furthermore, since input and weight alignments are performed independently, aligned inputs can be broadcast across output channels, thereby reducing the cost of alignment hardware.

C. SM-Based Assembled Column Addition Dataflow

Building upon the temporally cyclic-aligned input feeding and spatially cyclic-aligned weight mapping illustrated in Figs. 5 and 6 shows the proposed dataflow, SACA, that is tightly coupled with STCA. SACA leverages two additional characteristics of STCA to enable low-latency, low-energy MAC operations. The first characteristic is the vertical mapping of inputs and weights. Because inputs and weights are mapped vertically on the cell array, all bit-wise PPs summed within an intra-LCA belong to a single element-wise multiplication. If they were mapped horizontally, the bit-wise PPs computed within an intra-LCA would correspond to element-wise multiplications from different input channels, making it impossible to use sign-magnitude format for both inputs and weights. In our case, where PPs for a single element-wise multiplication are summed within an intra-LCA, both inputs and weights can use sign-magnitude format, thereby reducing the number of bit-serial MAC operations and latency in the CA-CIM macro.

The second characteristic is that the results computed within an intra-LCA repeat periodically due to the temporal cyclic shifting of inputs. When using sign-magnitude format for both inputs and weights, inputs are typically fed such that PPs of a single element-wise multiplication form a parallelogram shape, allowing PPs of identical bit-significance to be summed together. However, due to the cyclic shifting nature of input feeding in STCA, values for the LBS PP columns can be computed in earlier cycles, reducing computation latency, as shown in Fig. 6. In this case, PPs with different bit-significances must be summed simultaneously within an intra-LCA. To efficiently support this, each LCA is equipped with an MFA, and each LCA column contains both an LPAT and an HPAT. This arrangement delays the accumulation point of LBS PPs and HBS PPs as late as possible, minimizing adder tree overhead. Fig. 7 illustrates the detailed hardware implementation of the CIB and an MFA within the CA-CIM macro. The CIB consists of D flip-flops that cyclically shift input bits on each cycle, with three configurable modes: mode 0 for the sign and

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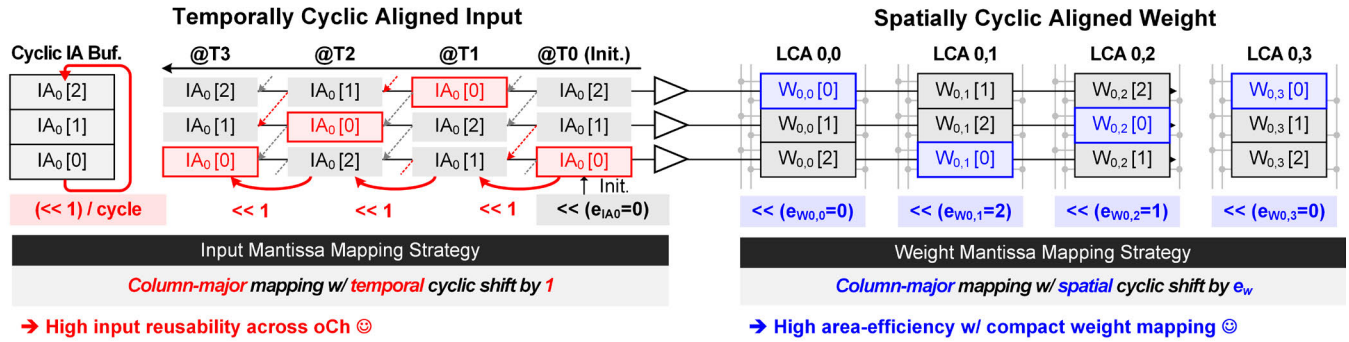


Fig. 5. Operation flow of STCA.

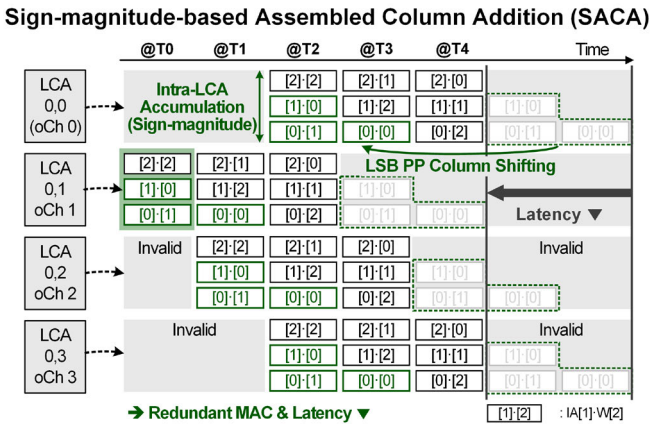


Fig. 6. Operation flow of SACA.

exponent bits, mode 1 for the MXFP4 mantissa, and mode 2 for the MXFP6 mantissa. The CIB feeds inputs to LCAs through the row decoder and WL driver, where each LCA contains 32 8T SRAM cells connected to three WLs and BLs. The MFA receives the sense amplifier outputs from three BLs and performs intra-LCA accumulation using SM format. The VRDU computes TAO by subtracting the weight exponent from L-CNT, generating start, end, and valid signals that control the gating mechanisms. The MFA output is converted from sign-magnitude to two's complement format before being forwarded to the LPAT and HPAT. This architecture enables PPs with different bit-significances to be summed simultaneously within an intra-LCA, thereby supporting low-latency SACA computation. It is important to note that the primary objective of STCA and SACA is not to reduce the latency of a single element-wise multiplication. Since STCA's effective mantissa alignment range is determined by the sum of the input activation exponent and the weight exponent, the cycle count per element can increase. However, SACA mitigates this overhead by shifting LBS PP columns into earlier cycles, and the early stopping mechanism in Section IV provides additional latency reduction. Consequently, while the per-element latency may be slightly longer, the compact weight mapping and sign-magnitude-based computation enable more operations to be computed within a single macro, leading to higher overall matrix multiplication throughput.

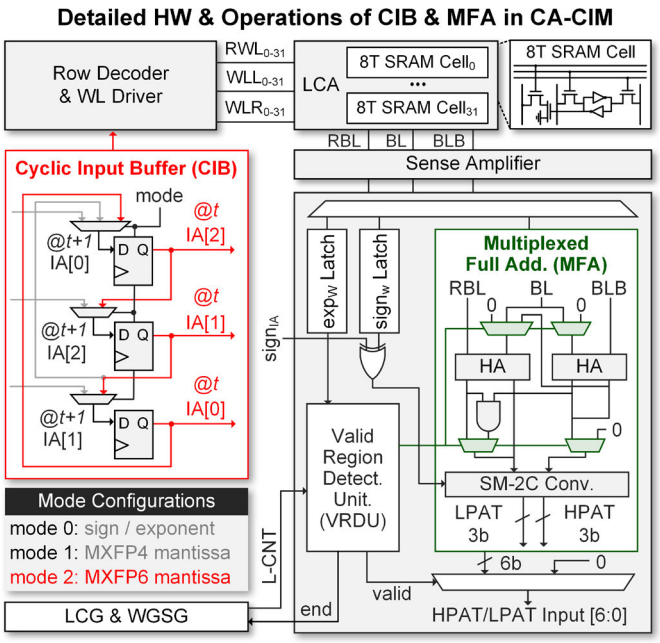


Fig. 7. Implementation of CIB and MFA for supporting STCA and SACA.

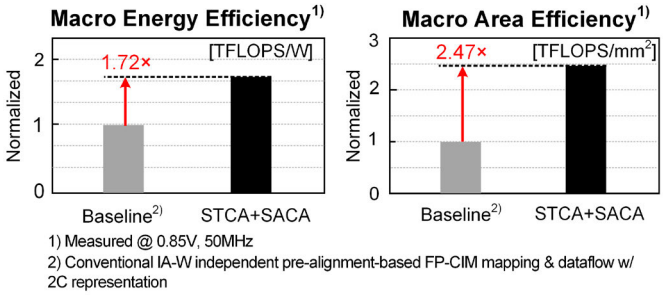


Fig. 8. Performance of STCA and SACA.

Fig. 8 shows the performance of STCA and SACA. By employing the tightly coupled mantissa alignment method and the dataflow of STCA and SACA, we achieved 1.72× higher macro energy efficiency and 2.47× higher macro area efficiency than the baseline [1].

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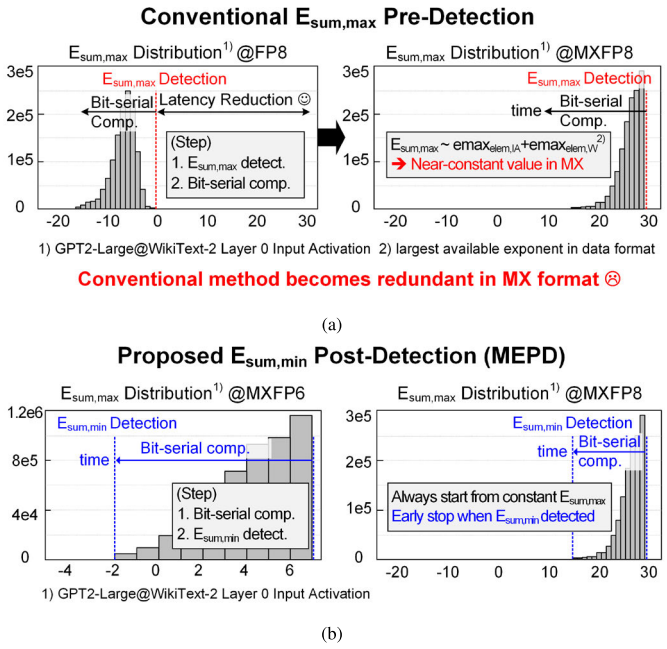


Fig. 9. Motivation of MEPD. (a) Conventional maximum exponent detection method. (b) Proposed MEPD.

IV. MX-Specialized FP Processing Pipeline

A. Motivation: Redundant $E_{Sum,Max}$ Detection in MX

In conventional FP operations, the maximum value of the exponent sum is typically pre-detected to reduce latency [4], [10], [14]. However, in the MX format, we found that the maximum exponent sum consistently stays near its theoretical upper bound due to its quantization characteristics, as shown in Fig. 9(a). According to the OCP MX Specification [9], the shared exponent is defined as in (1), where *ia* denotes the input activations within an MX block and $e_{\max,\text{elem}}$ represents the exponent of the largest normal number in the element data format

$$e_{\text{shared}} = \left\lfloor \log_2 \left(\max_{\text{i}} \left(|ia_{\text{i}}| \right) \right) \right\rfloor - e_{\max,\text{elem}}. \tag{1}$$

The exponent of a quantized element after MX quantization can be expressed as (2), where ia_{q} denotes the MX-quantized input activations

$$\begin{aligned} e_{\text{ia}_{\text{q}},\text{i}} &= e_{\text{ia,i}} - e_{\text{shared}} \\ &= e_{\max,\text{elem}} + e_{\text{ia,i}} - \left\lfloor \log_2 \left(\max_{\text{i}} \left(|ia_{\text{i}}| \right) \right) \right\rfloor \\ &= e_{\max,\text{elem}} + e_{\text{ia,i}} - e_{\text{ia,max}}. \end{aligned} \tag{2}$$

Then, the maximum value of the quantized element's exponent can be derived as (3), which equals $e_{\max,\text{elem}}$—a constant for a given MX format

$$e_{\text{ia}_{\text{q}},\max} = \max_{\text{i}} \left(e_{\text{ia}_{\text{q}},\text{i}} \right) = e_{\max,\text{elem}}. \tag{3}$$

In addition, we observe that the maximum value of the exponent sum closely approximates the sum of the maximum input and weight exponents, as shown in (4). Since this sum is a constant for a given MX format, the pre-detection of the maximum exponent sum becomes largely unnecessary

$$e_{\text{sum}_{\text{q}},\max} \approx e_{\text{ia}_{\text{q}},\max} + e_{\text{w}_{\text{q}},\max} = 2e_{\max,\text{elem}}. \tag{4}$$

Based on this observation, we propose an MX-specialized FP processing pipeline as shown in Fig. 9(b). Instead of detecting the maximum exponent sum at the beginning, we detect the minimum exponent sum at the end of the operation using a lightweight hardware module. This enables lossless early stopping, allowing computation to terminate earlier without any accuracy loss.

B. Online Valid Region Detection

To ensure the integrity of the computational output using the STCA and SACA methodologies discussed in Section III, gating must be applied during intervals when the LCAs generate invalid PPs. In conventional FP-supporting CIMs, pre-detecting the maximum exponent sum and aligning the input mantissa based on this value can be regarded as a process for ensuring valid computation results [4], [10], [14]. Instead of performing alignment in advance to ensure valid results, we employ an online valid region detection method that determines validity on-the-fly during each bit-serial operation. This approach effectively amortizes the conventional high-cost alignment operation over time. For online valid region detection, each CIM macro includes a global counter. The global counter is a 4-bit binary down counter whose output (G-CNT) is initialized to 0 and decrements by 1 each cycle. G-CNT represents the bit-significance of the PP columns currently being computed in each LCA of the CIM macro. By comparing G-CNT with the exponent sum of the input and weight in each LCA, the validity of the PP column computed in that cycle can be determined. We define the temporal alignment offset (TAO) as the difference between G-CNT and the exponent sum as shown in (5), where cnt_{global} denotes the G-CNT and e_{sum} denotes the exponent sum

$$tao_{\text{i}} = cnt_{\text{global}} - e_{\text{sum}}. \tag{5}$$

Then, the criterion for determining whether an LCA computation result is valid in a given cycle can be defined as in (6), where $mant_{\text{bit}}$ denotes the bit-width of the mantissa, which equals 1 for MXFP4 (E2M1) and 2 for MXFP6 (E3M2)

$$0 \le tao_{\text{i}} \le mant_{\text{bit}}. \tag{6}$$

C. Tao Reformulation

To further reduce hardware overhead, we reformulate the TAO calculation by leveraging the broadcasting nature of input activations across LCA columns. Instead of performing the summation of input activation exponent with the weight exponent within each LCA, we first subtract the input activation exponent from the G-CNT to generate a local counter value (L-CNT) as shown in (7). Here, cnt_{global}, cnt_{local}, e_{ia}, and $e_{\text{w,i}}$ denote G-CNT, L-CNT, the input activation exponent, and the weight exponent for output channel *i*, respectively.

$$tao_{\text{i}} = cnt_{\text{global}} - \left(e_{\text{ia}} + e_{\text{w,i}} \right)$$

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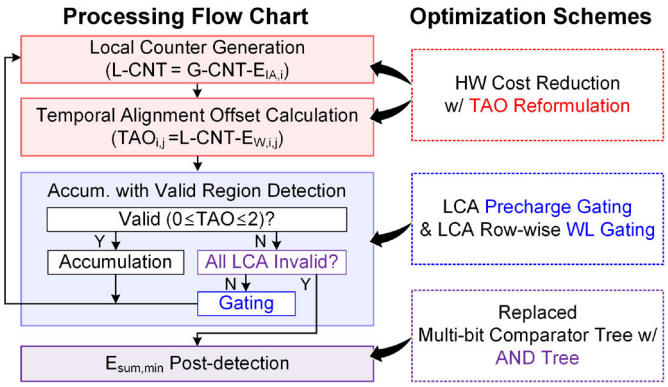


Fig. 10. Flowchart of MX-specialized FP processing pipeline and its optimization schemes.

$$= \left(cnt_{\text{global}} - e_{\text{ia}}\right) - e_{\text{w,i}} \\ = cnt_{\text{local}} - e_{\text{w,i}}. \quad (7)$$

L-CNT is generated for each LCA row, and the generated L-CNT is broadcast to different LCA columns. Each LCA computes TAO by calculating the difference between the broadcast L-CNT and the stored weight exponent. Through this reformulation, instead of performing exponent summation of input and weight followed by subtraction from the global counter at each LCA, TAO can be computed with a single subtraction per LCA row to obtain L-CNT, followed by a lower-bit subtraction between L-CNT and the weight exponent at each LCA.

D. Detailed MXFP Operation Flow

Fig. 10 illustrates the MX-specialized FP processing flowchart with TAO reformulation (TAOR) applied, along with the optimization schemes employed at each stage. Fig. 11 shows the micro-architecture within the CA-CIM macro corresponding to each stage of the MX-specialized FP processing pipeline. As shown in Fig. 10, the proposed pipeline consists of four steps.

1) Local Counter Generation: G-CNT from the 4-bit global counter in each CA-CIM macro is broadcast to the LCG in each LCA row. The LCG computes L-CNT by subtracting the 3-bit input exponent from the 4-bit G-CNT using a 4-bit subtractor. Both G-CNT and the input exponent are unsigned values, and the 3-bit input exponent is zero-padded at the MSB to form a 4-bit value before subtraction. Unlike a conventional 4-bit subtractor, since the input exponent spans only 3 bits, the FA computing the MSB does not need to generate a carry and is implemented using two XOR gates instead. The computed L-CNT is not only broadcast to LCAs in the same row but also checked within the LCG to detect whether the 4-bit L-CNT equals $e_{\max} + m_{\text{bit}}$, where $e_{\max}$ and m_{bit} denote the maximum exponent and the mantissa bit-width, respectively. The result is latched in a D flip-flop with a reset. This latched signal serves as an enable signal for clock gating, and the gated clock is broadcast to LCAs in the same row as the clock for D flip-flops that latch the start point of each LCA's valid region. This optimization exploits the constraint that the valid region lies within a limited range determined by the input exponent. Each LCA begins determining valid region status only when G-CNT meets this condition. Consequently, only the lower 3 bits of the 4-bit L-CNT need to be broadcast to LCAs in the same row, reducing the 4-bit subtraction previously required at each LCA for valid region determination to a 3-bit subtraction.

2) Tao Calculation: The 3-bit L-CNT broadcast to each LCA is subtracted from the latched weight exponent in the VRDU at each LCA to compute TAO.

3) Valid Region Detection: Since the mantissa bit-width varies depending on whether the input and weight precision are MXFP4 or MXFP6, the TAO value indicating the end of the valid region also differs. Precision information is received as a 1-bit mode signal and used to generate the end signal. For MXFP6, including the hidden bit, the mantissa bit-width is 3; thus, the valid region start condition is when TAO equals 0, and the end condition is when TAO equals 3. For MXFP4, the mantissa bit-width, including the hidden bit, is 2, so the start and end conditions correspond to TAO values of 0 and 2, respectively. Both the start and end signals remain at 1 once their respective conditions are satisfied until reset. The valid region occurs when the start signal is 1, and the end signal is 0, at which point the valid signal becomes 1. The valid signal is utilized for three types of gating. First, for intra-LCA pre-charge gating, the valid signal is ANDed with the corresponding LCA's pre-charge enable signal to perform pre-charge gating during invalid periods. Second, LCA row-wise WL gating uses the WGSG to gate the WL when all LCAs in an LCA row are in the invalid region, reducing power consumption. Third, LCA column-wise adder-tree gating ensures correct computation results by forwarding the PP accumulation results from the MFA to the LPAT and HPAT only when in the valid region.

4) $E_{Sum,Min}$ Post-Detection: Fig. 12 illustrates the timing chart of MXFP processing in CA-CIM with the proposed gating mechanisms. As the final step of the MX-specialized FP processing pipeline, the MEPU aggregates the end signals from all LCAs in the CIM macro through an AND-tree and detects the point at which the output becomes 1. This point indicates the cycle following the computation of the PP column with bit-significance corresponding to the minimum exponent summation value among all input-weight pairs computed in the CIM macro. At this point, the CIM macro triggers early stopping to reduce total computation latency.

We evaluated the proposed MX-specialized FP processing pipeline against a baseline that employs conventional maximum exponent sum pre-detection. As shown in Fig. 13, TAOR alone reduces power and area by 11.3% and 8.7%, respectively, by simplifying the per-LCA exponent computation. With the addition of MEPD, the total power and area reductions reach 68.8% and 63.2%, respectively, primarily due to the elimination of redundant maximum exponent detection logic and the lightweight valid region detection. Furthermore, MEPD enables early stopping, achieving a latency reduction of 22.5% under MXFP6. These improvements are primarily due to the elimination of redundant maximum exponent detection logic, the lightweight TAO-based valid region detection, and

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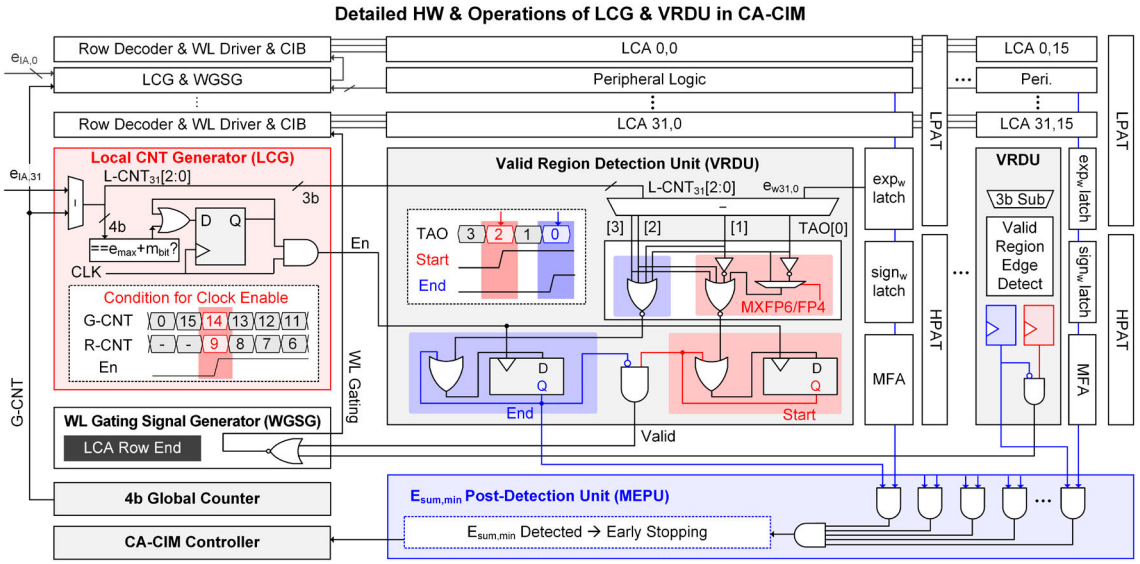


Fig. 11. Implementation of TAOR and MEPD.

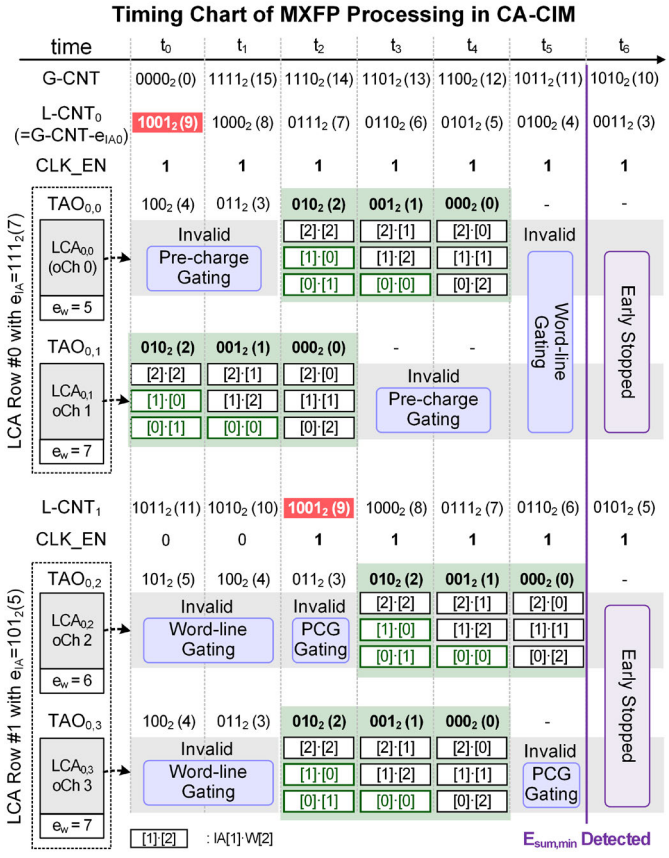


Fig. 12. Operation flow of MEPD with gatings.

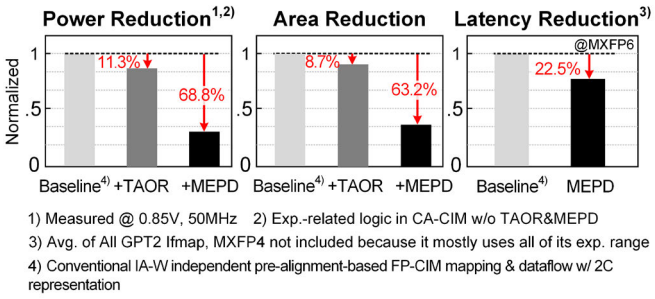


Fig. 13. Performance of TAOR and MEPD.

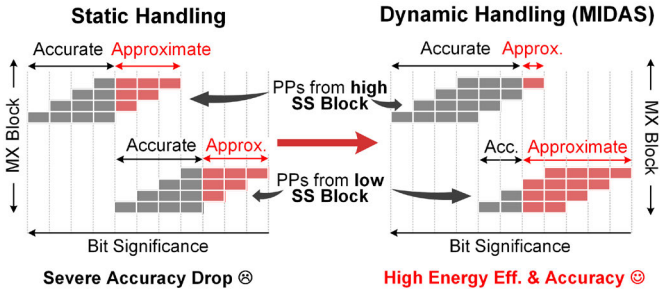


Fig. 14. Motivation of SADB.

the early-stopping mechanism enabled by minimum exponent post-detection.

V. MX Shared Scale-Aware Approximation

A. Motivation

Fig. 14 shows the motivation for shared scale-aware dynamic bit-significance handling. In the MX format, each block has an SS to adjust the dynamic range of its elements. Matrix multiplication between MX-quantized inputs and weights involves intra-block multiply-and-accumulation between inputs and weights, followed by inter-block accumulation that accounts for shared scales. Even if PPs have the same bit-significance within their respective intra-block multiply-and-accumulation operations, they may have different bit-significances during inter-block accumulation when the shared scales differ between blocks.

Several previous works have applied approximation to the accumulation of PPs with lower bit-significance. This approach is less aggressive than skipping computations for

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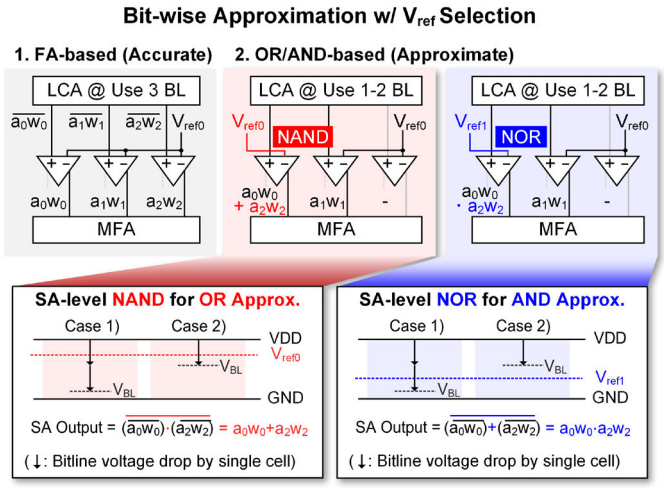


Fig. 15. Concept and implementation of SADB.

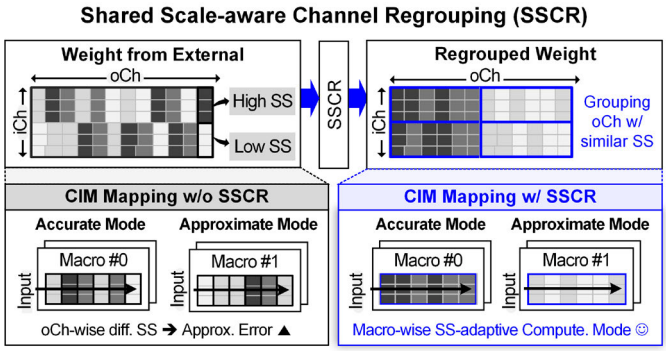


Fig. 16. Concept of SSCR.

lower bit-significance portions, effectively improving energy efficiency while minimizing accuracy loss. However, most prior studies used dedicated hardware logic, such as OR gates, for approximation, resulting in fixed bit positions for approximation [1], [5], [10], [11]. Such static bit-significance handling works well for integer formats or FP operations with pre-alignment. However, in the MX format, each block has its own shared scale. If approximation is applied at the same bit-positions across all blocks, intra-MAC results from inputs and weights with high shared scales will cause severe accuracy degradation during inter-block accumulation due to approximation applied to PPs with high bit-significance. Therefore, supporting dynamic bit-significance handling is essential for effectively applying approximation to MX-quantized matrix multiplication.

B. SADB-Significance Handling

To address this, we propose SADB-significance handling, which performs approximation by adjusting the reference voltage of the sense amplifier without requiring dedicated hardware logic. Since SADB can determine whether to apply approximation on a bit-wise basis, the range of bit significance for approximation can be flexibly adjusted for each block's shared scale. As explained in Fig. 15, for accurate computation, only one cell value is connected to a single BL. For approximate computation, two cells are connected in parallel, enabling OR or AND approximation depending on the sense amplifier's reference voltage level. The detailed operation is as follows: when the input drives the WL, the inverted weight (weight bar) connects to the pre-charged BL (initially at logic 1), effectively performing a NAND operation between the input bit and the weight bit. Using the reduced BL voltage level when one cell value is connected to a single BL as a reference, a reference voltage higher than this level yields an OR approximation from the sense amplifier, while a reference voltage lower than this level yields an AND approximation. Since bit sparsity approaches 50% at lower bit-positions where approximation is applied, accuracy loss can be minimized by alternating between AND and OR approximation across input channels. It is worth noting that SADB's robustness under PVT variations is fundamentally more favorable than that of conventional analog CIM. While analog CIM requires multi-level sensing with tightly spaced voltage margins, SADB performs only a binary sensing decision using two reference voltage levels. The two states to be distinguished differ by the BL voltage drop of a single cell, which provides a substantially wider sensing margin than the fine-grained levels required in analog CIM. Moreover, since SADB targets only LBS PP columns, any residual sensing error has a negligible impact on the final computation accuracy. The alternation of AND and OR approximations across input channels further provides statistical error cancellation against systematic reference voltage bias under PVT variations.

C. Shared Scale-Aware Channel Regrouping

Fig. 16 shows the concept of shared SSCR. SSCR is devised to improve the trade-off curve between energy efficiency and accuracy when using SADB. When the original MX-quantized weight tensor is directly mapped to the CIM, multiple output channels with different shared scales may be stored together in a single CIM macro. Since approximation is performed by adjusting the number of simultaneously driven WLs, all output channels mapped to a single CIM macro are forced to share the same approximation mode. Consequently, when output channels with different shared scales use the same approximation mode, the energy efficiency improvement from SADB can be diminished.

Fig. 17 illustrates the detailed hardware flow of SSCR in MXCE. SSCR uses the SS statistics of MX blocks in the weight tensor, which are known in advance, to classify weight MX blocks into high-SS and low-SS groups using SCRU. To ensure an equal number of blocks in both groups, only the median SS value of the MX blocks and the count of median-SS blocks that should belong to the low-SS group are provided. The incoming weight's SS is compared against these values to determine its group assignment. In the transpose buffer, weights from external memory along the output channel direction are stored in the corresponding buffer according to their classified group. Once classification is complete, the weights are transferred to the CIM macro. To avoid macro-level workload imbalancing that would occur if high-SS and low-SS blocks were mapped to different CIM macros, they

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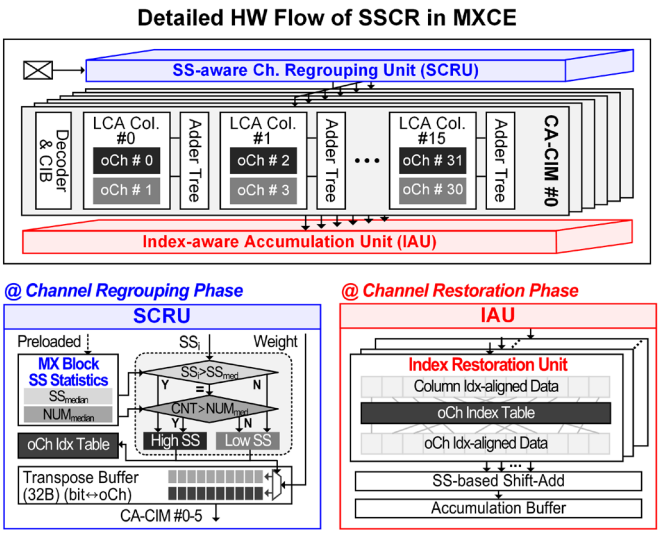


Fig. 17. Implementation of SSCR.

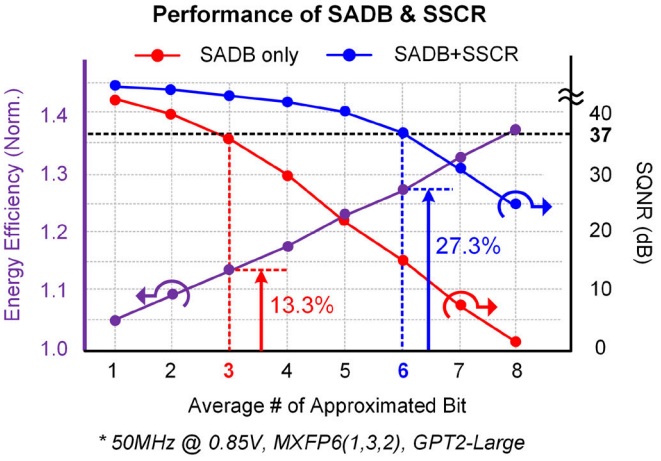


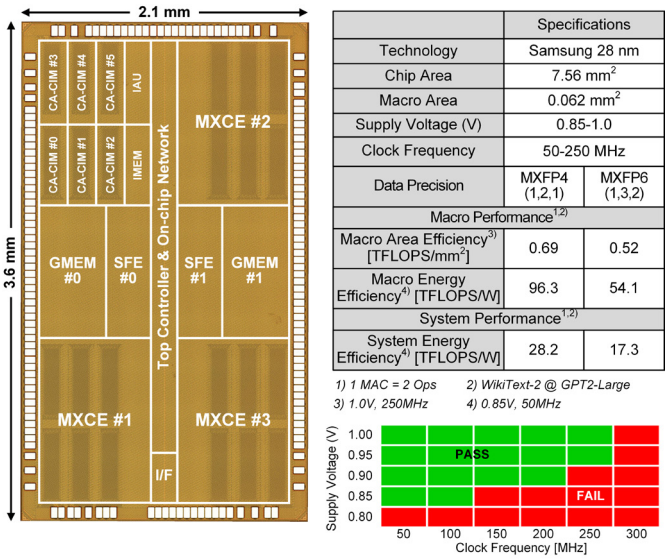
Fig. 18. Performance of SADB and SSCR.

are instead mapped to different rows within the same CIM macro. After the CIM macro computation completes, the IAU restores the output channel order and performs group-wise accumulation based on each block's SS.

We evaluated the effectiveness of combining SADB and SSCR against using SADB alone. As shown in Fig. 18, the results demonstrate that combining SADB and SSCR improves energy efficiency by 12.4% compared to SADB alone at 37 dB SQNR. Additionally, SADB with SSCR achieves up to 27.3% higher energy efficiency with an average of six approximated bits while maintaining acceptable SQNR levels. This improvement is attributed to the grouping of blocks with similar shared scales, which enables more aggressive approximation for low-SS blocks while preserving accuracy for high-SS blocks.

VI. Measurement Results

Fig. 19 shows the chip photograph and specifications of MIDAS. MIDAS is fabricated in Samsung 28-nm CMOS technology with a total die area of 7.56 mm^2 and a macro



	Specifications	
Technology	Samsung 28 nm	
Chip Area	7.56 mm^2	
Macro Area	0.062 mm^2	
Supply Voltage (V)	0.85-1.0	
Clock Frequency	50-250 MHz	
Data Precision	MXFP4 (1,2,1)	MXFP6 (1,3,2)
Macro Performance[1,2]		
Macro Area Efficiency[3] [TFLOPS/mm^2]	0.69	0.52
Macro Energy Efficiency[4] [TFLOPS/W]	96.3	54.1
System Performance[1,2]		
System Energy Efficiency[4] [TFLOPS/W]	28.2	17.3

1) 1 MAC = 2 Ops 2) WikiText-2 @ GPT2-Large
3) 1.0V, 250MHz 4) 0.85V, 50MHz

Fig. 19. Chip photograph with performance summary and shmoo plot.

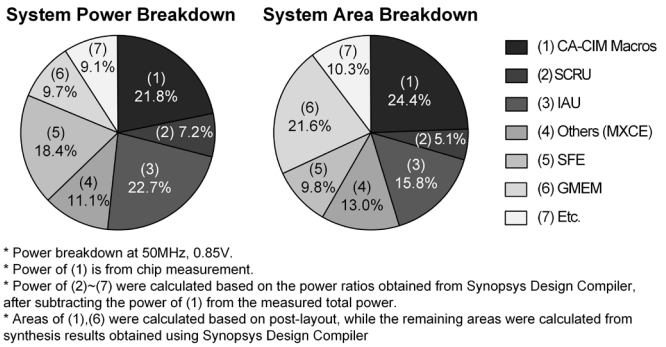


Fig. 20. Power and area breakdown of the MIDAS.

area of 0.062 mm^2. MIDAS operates at clock frequencies ranging from 50 to 250 MHz with a supply voltage of 0.85–1.0 V. The chip supports both MXFP4 (1,2,1) and MXFP6 (1,3,2) data precisions. Under MXFP4 at 1.0 V and 250 MHz, the measured macro area efficiency and macro energy efficiency are 0.69 TFLOPS/mm^2 and 96.3 TFLOPS/W, respectively. For MXFP6, these values are 0.52 TFLOPS/mm^2 and 54.1 TFLOPS/W. At 0.85 V and 50 MHz, the system energy efficiency reaches 28.2 TFLOPS/W for MXFP4 and 17.3 TFLOPS/W for MXFP6. The shmoo plot in Fig. 19 shows the pass/fail regions across the voltage–frequency operating space, confirming stable operation up to 250 MHz at 1.0 V and down to 0.85 V at 50 MHz.

Fig. 20 presents the system-level power and area breakdown of MIDAS. The CA-CIM macros account for 21.8% and 24.4% of the total chip power and area, respectively.

Fig. 21 details the macro-level power and area breakdown of a single CA-CIM macro. The MFA and adder tree dominate the macro power at 44.8%. The cell array occupies the largest share of the macro area (31.2%) but consumes only 11.7% of the power. The remaining components, such as the MEPU and global BL driver, contribute minimal overhead.

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TABLE I
PERFORMANCE COMPARISON TABLE

	ISSCC'23 [10]	ISSCC'23 [1]	ISSCC'24 [2]	ISSCC'24 [3]	ISSCC'25 [4]	**This Work**
Technology	22 nm	28 nm	28 nm	28 nm	16 nm	**28 nm**
MX Support	X	X	X	X	O	**O**
Target Model	CNN	CNN	CNN/Transformer	CNN/Transformer	CNN	**Transformer**
Target Task	Image Classification	Image Classification/ Detection/Segmentation	Image Classification/ Text Generation	Image Generation	Image Classification	**Generative AI**
Data Precision	BF16	BF16/INT8	POSIT16/ POSIT8	INT10/INT16/ FP16/BF16	INT8/BF16/ MXINT8	**MXFP4/MXFP6**
Die Area (mm^2)	18	0.146	1.41	3.67	8	**7.56**
Macro Area (mm^2)	18	0.146	NA	0.225 / 0.027	8	**0.062**
Supply Voltage (V)	0.6-0.8	0.6-0.9	0.55-1.0	0.6-1.0	0.4-0.8	**0.85 – 1.0**
Frequency (MHz)	NA	NA	0.55-1.0	50-540	NA	**50 – 250**
Macro Area Efficiency (TFLOPS/mm^2)	0.069-0.071	1.13-2.05	NA	NA	5.8[5]@BF16 11.4[5]@MXINT8	**0.69 @MXFP4[7]** **0.52 @MXFP6[7]**
Macro Energy Efficiency (TFLOPS/W)	16.2-70.2[1]	14-31.6@BFP16[2] 19.5-44@INT8[2]	16.3-83.2@POSIT8[2,3] 7.5-38.4@POSIT16[2,3]	80.3@FP16[4]	91.9[1,6]@BF16 133.5[1,6]@MXINT8	**96.3 @MXFP4[8]** **54.1 @MXFP6[8]**
System Energy Efficiency[9] (TFLOPS/W)	-	-	10.9-55.6@POSIT8[2,3] 5.4-27.6@POSIT16[2,3]	74.3@BF16[4] 67.9@FP16[4]	-	**28.2 @MXFP4[8]** **17.3 @MXFP6[8]**

1) Evaluated w/ 90% input sparsity 2) Evaluated w/ 50% input sparsity 3) 0.55V, 78MHz 4) 0.65V, 120MHz 5) 0.4V, 10% input toggle rate 6) 0.8V 7) 1.0V, 250MHz 8) 0.85V, 50MHz 9) EMA excluded

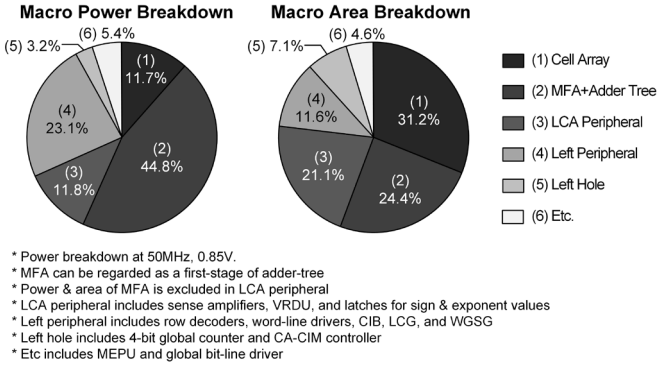


Fig. 21. Power and area breakdown of the CIM Macro.

TABLE II
BENCHMARK EVALUATION

	Large Language Model		
Model	GPT2-Large	Llama3.2-1B	Qwen2-1.5B
Dataset	WikiText-2	WikiText-2	WikiText-2
Parameter	774M	1B	1.5B
Precision[1](IA/W)	MXFP4 / MXFP4[2]	MXFP6 / MXFP6	MXFP6 / MXFP6
Accuracy (PPL)	20.11 (GT:19.44)	11.62 (GT: 10.99)	11.12 (GT: 10.70)
System Energy-Efficiency[3] (TFLOPS/W)	28.2	16.2	17.3

1) MXFP8 for attention block operations(QK^T, AV) 2) MXFP6 for projection layer in attention block 3) 0.85V, 50MHz

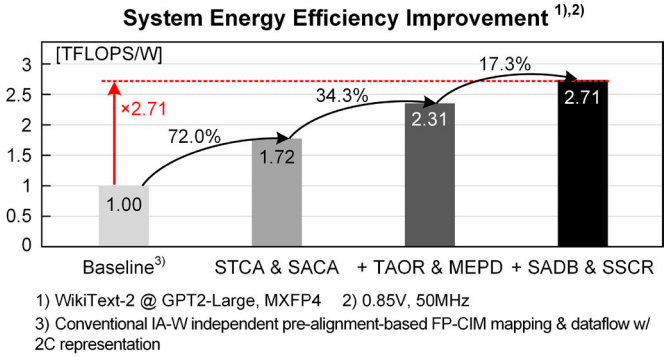


Fig. 22. Overall performance of key features.

Fig. 22 illustrates the system energy efficiency improvement contributed by each proposed feature, measured on WikiText-2 [15] with GPT2-Large using MXFP4 at 0.85 V and 50 MHz. Starting from the baseline, STCA and SACA provide a 72.0% improvement. Adding TAOR and MEPD yields an additional 34.3% gain. Finally, incorporating SADB and SSCR contributes a further 17.3% improvement. In total, MIDAS achieves 2.71× higher system energy efficiency compared to the baseline.

Table I compares MIDAS with state-of-the-art DCIM accelerators [1], [2], [3], [4], [10]. Unlike previous works that primarily target CNN models for image classification, MIDAS focuses on transformer-based GenAI applications with support for the sub-8-bit MX format. Also, MIDAS achieves 1.58× higher system energy efficiency and 3.32× higher macro energy efficiency over the state-of-the-art DCIM [2]. Unlike [4], which supports MX but mainly focuses on MXINT, MIDAS targets MXFP optimization, enabling broader application of the MX format for accuracy-sensitive GenAI models.

Table II summarizes the evaluation of MIDAS on three LLMs: GPT2-Large, Llama3.2-1B, and Qwen2-1.5B, all evaluated on the WikiText-2 dataset [15]. The measured perplexity values are 20.11, 11.62, and 11.12 for GPT2-Large, Llama3.2-1B, and Qwen2-1.5B, respectively, which are close to the ground truth values. The corresponding system energy

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efficiencies are 28.2 TFLOPS/W, 16.2 TFLOPS/W, and 17.3 TFLOPS/W.

VII. Conclusion

We propose MIDAS, a DCIM-based accelerator optimized for GenAI models leveraging the sub-8-bit MX data format. MIDAS introduces STCA for compact weight mapping and high input reusability, along with SACA for low-latency computation. The MX-aware MEPD eliminates redundant maximum exponent detection, enabling a lightweight exponent processing pipeline. Furthermore, SADB-significance handling combined with shared SSCR adapts approximation to block-wise dynamic bit-significance, improving energy efficiency while minimizing accuracy loss. Fabricated in 28-nm CMOS technology, MIDAS achieves macro energy and area efficiencies of 96.3 TFLOPS/W and 0.69 TFLOPS/mm^2 under MXFP4, respectively. These features enable MIDAS to achieve 1.58× higher energy efficiency on GPT-2 compared to prior DCIM [2], demonstrating its effectiveness as an MX-optimized AI accelerator for next-generation GenAI applications.

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